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Precalculus formula sheet

The Precalculus formulas with their conditions of validity: logic, powers and roots, special products and Ruffini's rule, equations and inequalities of every type, logarithms, exponentials, trigonometry, lines and conic sections.

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Logic and quantifiers

The truth tables and the examples are in the chapter Elementary logic.

Truth table of the connectives

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \Rightarrow q$
T	T	F	T	T	T
T	F	F	F	T	F
F	T	T	F	T	T
F	F	T	F	F	T

The disjunction $p \vee q$ is inclusive, that is, it is true also when p and q are both true. The implication is false *only if* the hypothesis is true and the conclusion is false.

Double implication

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$

The double implication is true when p and q have the same truth value. The symbol $\equiv$ indicates that two propositions have the same truth table.

De Morgan's laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q, \qquad \neg(p \vee q) \equiv \neg p \wedge \neg q$$

Negation exchanges “and” with “or” and negates the individual parts. For example the negation of $x > 0 \wedge x < 5$ is $x \leq 0 \vee x \geq 5$.

Implication, negation and contrapositive

$$p \Rightarrow q \equiv \neg p \vee q, \qquad \neg(p \Rightarrow q) \equiv p \wedge \neg q,$$

$$p \Rightarrow q \equiv \neg q \Rightarrow \neg p.$$

The contrapositive $\neg q \Rightarrow \neg p$ is equivalent to the original implication. The converse $q \Rightarrow p$ has a different truth table, therefore it can be false even when $p \Rightarrow q$ is true.

Necessary and sufficient condition

$$p \Rightarrow q \quad \Longleftrightarrow \quad p \text{ is sufficient for } q \text{ and } q \text{ is necessary for } p$$
$$p \Leftrightarrow q \quad \Longleftrightarrow \quad p \text{ is necessary and sufficient for } q$$

From the contrapositive it follows that, if the necessary condition q is false, then p is also false. For example "n multiple of 4" is sufficient for "n even", while "n even" is necessary for "n multiple of 4".

Negation of quantifiers

$$\neg\big(\forall x \in A : p(x)\big) \equiv \exists x \in A : \neg p(x),$$
$$\neg\big(\exists x \in A : p(x)\big) \equiv \forall x \in A : \neg p(x).$$

With several quantifiers all of them are exchanged, from the outermost to the innermost, and finally the predicate is negated. For example the negation of $\forall x\ \exists y : y > x$ is $\exists x\ \forall y : y \leq x$. A single counterexample is enough to make a universal proposition false.

Proof schemes

direct:	$p \Rightarrow \cdots \Rightarrow q$
by contrapositive:	$\neg q \Rightarrow \neg p$
by contradiction:	$p \wedge \neg q \Rightarrow$ contradiction

The three schemes prove the same implication $p \Rightarrow q$. Proof by contradiction is based on the equivalence $\neg(p \Rightarrow q) \equiv p \wedge \neg q$.

Powers

The definitions and the examples are in the chapter Powers and roots.

Zero and negative exponents

$$a^0 = 1, \qquad a^{-n} = \frac{1}{a^n}, \qquad \left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$$

They hold for $a \neq 0$, $b \neq 0$ and n a natural number. The expression 0^0 has no meaning. For example $2^{-3} = \dfrac{1}{8}$ and $\left(\dfrac{2}{3}\right)^{-1} = \dfrac{3}{2}$.

Power with a negative base

$$(-a)^n = \begin{cases} a^n & \text{if } n \text{ is even} \\ -a^n & \text{if } n \text{ is odd} \end{cases}$$

It holds for every real a and natural n. The expression -2^4 denotes the opposite of 2^4, that is, -16, while $(-2)^4 = 16$.

Properties of powers

$$a^m \cdot a^n = a^{m+n}, \qquad \frac{a^m}{a^n} = a^{m-n}, \qquad (a^m)^n = a^{mn},$$
$$a^n \cdot b^n = (ab)^n, \qquad \frac{a^n}{b^n} = \left(\frac{a}{b}\right)^n.$$

They hold for $a, b > 0$ and real exponents. With negative bases they hold for integer exponents, where the powers are defined.

Warning

The properties concern products and quotients. For a sum, in general, $(a+b)^n \neq a^n + b^n$ and $a^m + a^n \neq a^{m+n}$. For example $(3+4)^2 = 49$ and $3^2 + 4^2 = 25$; likewise $2^2 + 2^3 = 12$ and $2^5 = 32$.

Power with a rational exponent

$$a^{\frac{m}{n}} = \sqrt[n]{a^m}, \qquad a^{\frac{1}{n}} = \sqrt[n]{a}, \qquad a^{-\frac{m}{n}} = \frac{1}{\sqrt[n]{a^m}}$$

They hold for $a > 0$, m an integer and n a natural number with $n \geq 2$. With this definition the properties of powers continue to hold, for example $\sqrt{2} \cdot \sqrt[3]{2} = 2^{\frac{1}{2}+\frac{1}{3}} = 2^{\frac{5}{6}}$.

Comparing powers

$$x < y \iff a^x < a^y \qquad (a > 1),$$
$$x < y \iff a^x > a^y \qquad (0 < a < 1),$$
$$0 < a < b \implies a^x < b^x \qquad (x > 0).$$

With exponent $x < 0$ the third inequality is reversed. Powers with different bases and exponents are compared after rewriting them with the same base or with the same exponent, for example $2^{30} = 8^{10} < 9^{10} = 3^{20}$.

Roots and rationalization

The conditions of existence and the three cases of rationalization are explained in the same chapter Powers and roots.

n-th root and conditions of existence

$$\sqrt[n]{a} = x \iff x^n = a \qquad (n \text{ odd},\ a \in \mathbb{R}),$$

$$\sqrt[n]{a} = x \iff \begin{cases} x \geq 0 \\ x^n = a \end{cases} \qquad (n \text{ even},\ a \geq 0).$$

With an even index the root exists *only if* the radicand is greater than or equal to zero, and its value is greater than or equal to zero. For example $\sqrt{49} = 7$, while the equation $x^2 = 49$ has the solutions $x = \pm 7$; with an odd index $\sqrt[3]{-8} = -2$.

Properties of roots

$$\sqrt[n]{ab} = \sqrt[n]{a}\,\sqrt[n]{b}, \qquad \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}},$$

$$\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}, \qquad \sqrt[n]{a^k} = \left(\sqrt[n]{a}\right)^k.$$

They hold for $a \geq 0$ and $b \geq 0$ ($b > 0$ in the quotient), with indices $m, n \geq 2$ and natural exponent $k \geq 1$. If the indices are odd they hold for all real a and b, with $b \neq 0$ in the quotient.

Invariance property and comparing roots

$$\sqrt[n]{a^k} = \sqrt[np]{a^{kp}}, \qquad 0 \leq a < b \implies \sqrt[n]{a} < \sqrt[n]{b}$$

They hold for $a \geq 0$ and $p \geq 1$ a natural number. Roots with different indices are brought to the common index given by the least common multiple of the indices. For example $\sqrt{3} = \sqrt[6]{27}$ and $\sqrt[3]{5} = \sqrt[6]{25}$, therefore $\sqrt{3} > \sqrt[3]{5}$.

Root of a square and moving factors under the root sign

$$\sqrt{a^2} = |a|, \qquad \sqrt{a^2 b} = |a|\sqrt{b}, \qquad a\sqrt{b} = \sqrt{a^2 b} \quad (a \geq 0)$$

The first holds for every real a, the second and the third for $b \geq 0$. For $a < 0$ we have $a\sqrt{b} = -\sqrt{a^2 b}$, for example $-2\sqrt{3} = -\sqrt{12}$. With an even index n, $\sqrt[n]{a^n} = |a|$; with an odd index, $\sqrt[n]{a^n} = a$.

Sum of like radicals

$$a\sqrt[n]{c} + b\sqrt[n]{c} = (a + b)\sqrt[n]{c}$$

It holds for radicals with the same index and the same radicand, with $c \geq 0$ if n is even. First the factors are taken out of the root sign, for example $\sqrt{12} + \sqrt{27} = 2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$.

Rationalization

$$\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}, \qquad \frac{1}{\sqrt[n]{a^k}} = \frac{\sqrt[n]{a^{n-k}}}{a},$$
$$\frac{1}{\sqrt{a} \pm \sqrt{b}} = \frac{\sqrt{a} \mp \sqrt{b}}{a - b}.$$

They hold for $a > 0$, with $0 < k < n$ in the second and $b \geq 0$, $a \neq b$ in the third. The third multiplies by the conjugate binomial, since $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$; with a number in place of a root we have for example $\frac{1}{\sqrt{5} - 1} = \frac{\sqrt{5} + 1}{4}$.

Special products

The expansions and their use in factorization are in the chapter Factorization, Ruffini's rule and polynomial division.

Square of a binomial and of a trinomial

$$(A + B)^2 = A^2 + 2AB + B^2, \qquad (A - B)^2 = A^2 - 2AB + B^2,$$
$$(A + B + C)^2 = A^2 + B^2 + C^2 + 2AB + 2AC + 2BC.$$

They hold for every value of A, B, C. Read from right to left they factor a trinomial made of two squares and twice the product of the bases, for example $x^2 - 10x + 25 = (x - 5)^2$.

Cube of a binomial

$$(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3,$$
$$(A - B)^3 = A^3 - 3A^2B + 3AB^2 - B^3.$$

They hold for every value of A and B. For example $(x + 2)^3 = x^3 + 6x^2 + 12x + 8$.

Difference of squares

$$A^2 - B^2 = (A - B)(A + B)$$

It holds for every value of A and B and can be applied several times in a row, for example $x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1)$.

Sum and difference of cubes

$$A^3 + B^3 = (A + B)(A^2 - AB + B^2),$$
$$A^3 - B^3 = (A - B)(A^2 + AB + B^2).$$

They hold for every value of A and B. The second factor, called the false square, as a trinomial in A has discriminant $-3B^2$, which is negative for $B \neq 0$; therefore it has no real roots and the factorization ends.

Sum and difference of powers

$$A^n - B^n = (A - B)\left(A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}\right),$$
$$A^n + B^n = (A + B)\left(A^{n-1} - A^{n-2}B + \cdots - AB^{n-2} + B^{n-1}\right).$$

The first holds for every natural number $n \geq 2$, the second for n *odd*. For $n = 3$ one recovers the difference and the sum of cubes.

Power of a binomial and Pascal's triangle

$$(A + B)^n = \sum_{k=0}^{n} \binom{n}{k} A^{n-k}B^k, \qquad \binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

It holds for n a natural number. The coefficients are the rows of Pascal's triangle, for example $(A + B)^4 = A^4 + 4A^3B + 6A^2B^2 + 4AB^3 + B^4$. In the expansion of $(A - B)^n$ the signs of the terms alternate.

Special trinomial

$$x^2 + (a + b)\,x + ab = (x + a)(x + b)$$

It is applied when the two numbers a and b can be found mentally. For example $x^2 - x - 12 = (x + 3)(x - 4)$, since $3 + (-4) = -1$ and $3 \cdot (-4) = -12$.

Factoring out a common factor and factoring by grouping

$$AB + AC = A\,(B + C),$$
$$AC + AD + BC + BD = A\,(C + D) + B\,(C + D) = (A + B)(C + D).$$

Factoring by grouping is completed when the groups leave the same factor, for example $x^3 + 2x^2 + 3x + 6 = x^2(x + 2) + 3(x + 2) = (x + 2)(x^2 + 3)$.

Warning

The sum of squares $x^2 + k^2$ with $k \neq 0$ does not factor into first-degree factors with real coefficients. For example $x^2 + 4 \geq 4$ for every x, therefore $x^2 + 4$ has no real roots.

Division, Ruffini's rule and factorization

The theorems with their proofs are in the chapter Factorization, Ruffini's rule and polynomial division.

Identity principle for polynomials

$$a_n x^n + \cdots + a_1 x + a_0 = b_n x^n + \cdots + b_1 x + b_0 \ \ \forall x \in \mathbb{R} \iff a_k = b_k \ \ \forall k$$

With the principle one determines unknown coefficients, for example from $x^2 + bx + c = x^2 - 4x + 3$ it follows that $b = -4$ and $c = 3$. In the product of two nonzero polynomials the degrees add, that is, $\deg(PQ) = \deg P + \deg Q$.

Division with remainder

$$\begin{cases} A(x) = B(x)\,Q(x) + R(x) \\ R(x) = 0 \ \text{ or } \ \deg R < \deg B \end{cases}$$

For every nonzero divisor B the quotient Q and the remainder R exist and are unique. If $\deg A \geq \deg B$, the quotient has degree $\deg A - \deg B$; if $R = 0$, the polynomial B divides A.

Remainder theorem and Ruffini's theorem

$$P(x) = (x - c)\,Q(x) + P(c),$$

$$(x - c) \text{ divides } P(x) \iff P(c) = 0.$$

The remainder of the division by $x - c$ is therefore $P(c)$ and is computed without carrying out the division. For a divisor $ax - b$ with $a \neq 0$ the remainder is $P\left(\dfrac{b}{a}\right)$.

Ruffini's scheme

	1	-2	-5	6
1		1	-1	-6
	1	-1	-6	0

In the first row the coefficients are written in decreasing order of degree, with 0 in place of the missing degrees; each number in the result row is multiplied by c and added to the coefficient in the next column. The last number is the remainder $P(c)$, the others are the coefficients of the quotient, of degree $\deg P - 1$. Here $x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 - x - 6)$.

Rational roots of a polynomial

$$P\left(\frac{p}{q}\right) = 0 \implies p \mid a_0 \ \text{ and } \ q \mid a_n$$

It holds for $P(x) = a_n x^n + \cdots + a_0$ with integer coefficients and $a_0 \neq 0$, and for $\frac{p}{q}$ in lowest terms. The rational roots are therefore sought among a finite number of candidates; if $a_n = 1$, they are integers that divide the constant term.

Factorization of the quadratic trinomial

$$ax^2 + bx + c = a\,(x - x_1)(x - x_2)$$

It holds for $a \neq 0$ and $\Delta = b^2 - 4ac \geq 0$, with x_1 and x_2 the roots of the trinomial; if $\Delta = 0$ we have $ax^2 + bx + c = a\,(x - x_1)^2$. If $\Delta < 0$ the trinomial does not factor into real first-degree factors.

Multiplicity of a root

$$P(x) = (x - c)^m\, Q(x), \qquad Q(c) \neq 0$$

In this case the root c has multiplicity $m \geq 1$. The graph of P crosses the x-axis at c if m is odd and touches it without crossing if m is even.

Linear and quadratic equations

The proof of the quadratic formula is in the chapter Polynomial and rational equations and inequalities.

Linear equation

$$ax + b = 0 \iff x = -\frac{b}{a}$$

It holds for $a \neq 0$. For $a = 0$ the equation has no solutions if $b \neq 0$ and is an identity if $b = 0$.

Quadratic formula

$$ax^2 + bx + c = 0 \iff x = \frac{-b \pm \sqrt{\Delta}}{2a}, \qquad \Delta = b^2 - 4ac$$

It holds for $a \neq 0$ and $\Delta \geq 0$. For $\Delta > 0$ the solutions are two and distinct, for $\Delta = 0$ there is only the double solution $x = -\frac{b}{2a}$, for $\Delta < 0$ the equation **has no real solutions**.

Reduced quadratic formula

$$x = \frac{-\frac{b}{2} \pm \sqrt{\frac{\Delta}{4}}}{a}, \qquad \frac{\Delta}{4} = \left(\frac{b}{2}\right)^2 - ac$$

It holds for $a \neq 0$ and $\frac{\Delta}{4} \geq 0$. It is the quadratic formula with numerator and denominator divided by 2, and it shortens the computations when b is even. For example $x^2 - 6x + 4 = 0$ has $\frac{\Delta}{4} = 9 - 4 = 5$, therefore $x = 3 \pm \sqrt{5}$.

Incomplete quadratic equations

$$ax^2 + bx = 0 \iff x = 0 \ \vee \ x = -\frac{b}{a},$$

$$ax^2 + c = 0 \iff x = \pm\sqrt{-\frac{c}{a}}.$$

They hold for $a \neq 0$. The second has real solutions *only if* $-\frac{c}{a} \geq 0$; for $-\frac{c}{a} < 0$ it has no solutions.

Sum and product of the roots

$$x_1 + x_2 = -\frac{b}{a}, \qquad x_1 \cdot x_2 = \frac{c}{a}$$

They hold for $a \neq 0$ and $\Delta \geq 0$, with x_1 and x_2 the roots of $ax^2 + bx + c = 0$. Consequently the two numbers with sum s and product p are the solutions of $x^2 - sx + p = 0$.

Zero product property

$$A \cdot B = 0 \iff A = 0 \ \vee \ B = 0$$

An equation $P(x) = 0$ of degree greater than two is solved by factoring P and setting each factor equal to zero. A polynomial of degree n has at most n real roots.

Biquadratic and trinomial equations

$$ax^4 + bx^2 + c = 0 \quad \xrightarrow{t = x^2} \quad at^2 + bt + c = 0$$

Every solution $t > 0$ gives $x = \pm\sqrt{t}$, the solution $t = 0$ gives $x = 0$, the solutions $t < 0$ are discarded. Likewise the equation $ax^{2n} + bx^n + c = 0$ is solved with the substitution $t = x^n$.

Polynomial and rational inequalities

The examples with the sign table are in the chapter Polynomial and rational equations and inequalities.

Equivalence principles for inequalities

$$A < B \iff A + C < B + C,$$

$$A < B \iff AC < BC \quad (C > 0), \qquad A < B \iff AC > BC \quad (C < 0).$$

Multiplying both sides by a negative quantity reverses the direction of the inequality. For example from $-2x > -4$ it follows that $x < 2$.

Sign of the quadratic trinomial

$$\begin{aligned}
\Delta > 0 &\implies ax^2 + bx + c > 0 \text{ for } x < x_1 \vee x > x_2 \\
\Delta = 0 &\implies ax^2 + bx + c > 0 \text{ for } x \neq x_0 \\
\Delta < 0 &\implies ax^2 + bx + c > 0 \text{ for every } x
\end{aligned}$$

They hold for $a > 0$, with $x_1 < x_2$ the roots of the trinomial and x_0 the double root; for $x_1 < x < x_2$ the trinomial is negative. For $a < 0$ the signs are the opposite, and alternatively one multiplies by -1 reversing the direction of the inequality.

Sign of a product and of a quotient

$$\operatorname{sign}\left(\frac{N(x)}{D(x)}\right) = \operatorname{sign}\big(N(x)\,D(x)\big) \qquad (D(x) \neq 0)$$

A product is positive where the number of negative factors is even and negative where it is odd; in the sign table each factor occupies one row. A factor $(x - c)^m$ with m even is positive for $x \neq c$, with m odd it has the sign of $x - c$.

Rational equations

$$\frac{N(x)}{D(x)} = 0 \iff \begin{cases} N(x) = 0 \\ D(x) \neq 0 \end{cases}$$

The conditions of existence $D(x) \neq 0$ are written for all the denominators of the original equation, before reducing to a common denominator. The solutions of $N(x) = 0$ that make a denominator vanish are discarded.

Rational inequalities

$$\frac{N(x)}{D(x)} > 0 \iff N(x)\,D(x) > 0,$$

$$\frac{N(x)}{D(x)} \geq 0 \iff \begin{cases} N(x)\,D(x) \geq 0 \\ D(x) \neq 0 \end{cases}$$

The zeros of the denominator are excluded for every direction of the inequality; with $\geq$ and $\leq$ the zeros of the numerator are included.

Warning

Multiplying both sides by a denominator presupposes its sign, which depends on x. From $\dfrac{1}{x} < 1$ it does not follow that $1 < x$, since $x = -1$ satisfies the first inequality and not the second. The solutions of $\dfrac{1}{x} < 1$ are $x < 0 \ \vee \ x > 1$.

Systems of inequalities

$$\begin{cases} A(x) > 0 \\ B(x) > 0 \end{cases} \iff x \in S_A \cap S_B$$

Here S_A and S_B are the solution sets of the individual inequalities. The solutions of the system are therefore the intervals common to all the inequalities.

Absolute value

Every solving form descends from the definition by cases, as shown in the chapter Equations and inequalities with absolute value.

Absolute value and distance

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

The absolute value is always greater than or equal to zero and equals zero *only if* $x = 0$. The number $|x - c|$ is the distance between x and c on the real line; for an expression A we have $|A| = A$ where $A \geq 0$ and $|A| = -A$ where $A < 0$.

Properties of the absolute value

$$|ab| = |a|\,|b|, \qquad \left|\frac{a}{b}\right| = \frac{|a|}{|b|}, \qquad |-a| = |a|,$$
$$|a|^2 = a^2, \qquad \sqrt{a^2} = |a|, \qquad |a+b| \leq |a| + |b|.$$

They hold for all real a, b, with $b \neq 0$ in the quotient. The last one is the **triangle inequality**, and equality holds if and only if $ab \geq 0$.

Equation |A| = k

$$|A| = k \iff A = k \ \vee \ A = -k \qquad (k > 0)$$

For $k = 0$ the equation is equivalent to $A = 0$. For $k < 0$ it has no solutions, because $|A| \geq 0$.

Equation |A| = |B|

$$|A| = |B| \iff A = B \ \vee \ A = -B$$

The formula holds without additional conditions, because both sides are greater than or equal to zero.

Equation |A| = B

$$|A| = B \iff \begin{cases} B \geq 0 \\ A = B \end{cases} \cup \begin{cases} B \geq 0 \\ A = -B \end{cases}$$

The left-hand side is not negative, therefore the equation imposes the condition $B \geq 0$. The solutions that do not satisfy it are discarded.

Inequalities |A| < k and |A| ≤ k

$$|A| < k \iff -k < A < k, \qquad |A| \leq k \iff -k \leq A \leq k$$

They hold for $k > 0$. For $k \leq 0$ the inequality $|A| < k$ has no solutions; the inequality $|A| \leq 0$ is equivalent to $A = 0$.

Inequalities |A| > k and |A| ≥ k

$$|A| > k \iff A < -k \ \vee \ A > k, \qquad |A| \geq k \iff A \leq -k \ \vee \ A \geq k$$

They hold for $k > 0$. For $k = 0$ we have $|A| > 0 \iff A \neq 0$. For $k < 0$ the inequality $|A| > k$ is satisfied on the whole domain of A, and the same holds for $|A| \geq k$ with $k \leq 0$.

Inequalities |A| < B and |A| > B

$$|A| < B \iff \begin{cases} A < B \\ A > -B \end{cases} \qquad |A| > B \iff A > B \ \vee \ A < -B$$

The two forms hold whatever the sign of B, because the system on the left by itself implies $B > 0$ and the union on the right includes the points with $B < 0$. With $\leq$ and $\geq$ the forms are the same, with non-strict inequalities.

Inequality |A| < |B|

$$|A| < |B| \iff A^2 < B^2 \iff (A - B)(A + B) < 0$$

It holds because both sides are greater than or equal to zero, therefore the comparison between the absolute values is equivalent to the comparison between the squares.

Several absolute values: case analysis by intervals

$$\begin{cases} x < 0 \\ -2x + 2 = 4 \end{cases} \cup \begin{cases} 0 \le x < 2 \\ 2 = 4 \end{cases} \cup \begin{cases} x \ge 2 \\ 2x - 2 = 4 \end{cases}$$

The arguments of $|x| + |x - 2| = 4$ vanish at 0 and at 2, which divide the line into three intervals; in each interval every absolute value is written without bars according to the sign of its argument. The solutions are the union of the results of the intervals, here $x = -1$ and $x = 3$.

Radical equations and inequalities

The reason for each condition is explained in the chapter Radical equations and inequalities.

Equation √A = B

$$\sqrt{A} = B \iff \begin{cases} B \ge 0 \\ A = B^2 \end{cases}$$

The condition of existence $A \ge 0$ follows from $A = B^2$. Without the condition $B \ge 0$ squaring introduces extraneous solutions: from $\sqrt{x} = x - 2$ one obtains $x^2 - 5x + 4 = 0$, but the root $x = 1$ gives $x - 2 = -1 < 0$, therefore the only solution is $x = 4$.

Equation √A = √B

$$\sqrt{A} = \sqrt{B} \iff \begin{cases} A \ge 0 \\ A = B \end{cases}$$

The condition $B \ge 0$ follows from $A = B$ and $A \ge 0$. In place of $A \ge 0$ one may impose $B \ge 0$, choosing the simpler radicand.

Inequalities √A < B and √A ≤ B

$$\sqrt{A} < B \iff \begin{cases} A \ge 0 \\ B > 0 \\ A < B^2 \end{cases} \qquad \sqrt{A} \le B \iff \begin{cases} A \ge 0 \\ B \ge 0 \\ A \le B^2 \end{cases}$$

The first condition is the existence of the root and the second follows from $\sqrt{A} \ge 0$. With both sides greater than or equal to zero, squaring preserves the direction of the inequality.

Inequalities √A > B and √A ≥ B

$$\sqrt{A} > B \iff \begin{cases} B < 0 \\ A \geq 0 \end{cases} \cup \begin{cases} B \geq 0 \\ A > B^2 \end{cases}$$

Where $B < 0$ it is enough that the root exists; in the second system the condition $A \geq 0$ follows from $A > B^2$. For $\sqrt{A} \geq B$ the second system becomes $B \geq 0$, $A \geq B^2$.

Inequality √A < √B

$$\sqrt{A} < \sqrt{B} \iff \begin{cases} A \geq 0 \\ A < B \end{cases}$$

The condition of existence $B \geq 0$ follows from $0 \leq A < B$.

Roots with odd index

$$\sqrt[3]{A} = B \iff A = B^3,$$

$$\sqrt[3]{A} < B \iff A < B^3, \qquad \sqrt[3]{A} > B \iff A > B^3.$$

They hold without conditions of existence and without cases on the sign of the sides, because $y = x^3$ is strictly increasing on all of $\mathbb{R}$. The same holds for every odd index n, with B^n in place of B^3.

Logarithms

The properties and the solving methods are explained in the chapter Logarithmic equations and inequalities.

Definition of logarithm

$$\log_a b = c \iff a^c = b$$

It holds for $a > 0$, $a \neq 1$ and $b > 0$. The logarithm exists *only* for positive arguments, because $a^c > 0$ for every c. The logarithm to base e is denoted by ln.

Identities and special values

$$\log_a 1 = 0, \qquad \log_a a = 1, \qquad \log_a(a^x) = x, \qquad a^{\log_a x} = x$$

They hold for $a > 0$, $a \neq 1$; the third for every real x, the fourth for $x > 0$. With the third a number is written as a logarithm, for example $2 = \log_3 9$.

Properties of logarithms

$$\log_a(xy) = \log_a x + \log_a y, \qquad \log_a \frac{x}{y} = \log_a x - \log_a y,$$
$$\log_a(x^p) = p \log_a x.$$

They hold for $a > 0$, $a \neq 1$, $x > 0$, $y > 0$ and p real. With the unknown in the arguments, the conditions of existence are written before applying them.

Logarithm of an even power

$$\log_a(x^2) = 2 \log_a |x|$$

It holds for $x \neq 0$. The equality $\log_a(x^2) = 2 \log_a x$ holds *only* for $x > 0$, therefore, applied to an equation, it can restrict its domain.

Change of base

$$\log_a x = \frac{\log_b x}{\log_b a}, \qquad \log_a b = \frac{1}{\log_b a}, \qquad \log_{a^k} x = \frac{1}{k} \log_a x$$

They hold for $a, b > 0$ different from 1, $x > 0$ and $k \neq 0$; for $k = -1$ one obtains $\log_{\frac{1}{a}} x = -\log_a x$. For example $\log_4 8 = \dfrac{\log_2 8}{\log_2 4} = \dfrac{3}{2}$.

Logarithmic function

$$y = \log_a x, \qquad x > 0$$

For $a > 1$ it is strictly increasing, for $0 < a < 1$ it is strictly decreasing; the graph passes through $(1, 0)$ and has the line $x = 0$ as a vertical asymptote. For $a > 1$ the logarithm is negative for $0 < x < 1$ and positive for $x > 1$.

Warning

The properties of logarithms concern products, quotients and powers. In general $\log_a(x + y) \neq \log_a x + \log_a y$ and $(\log_a x)^2 \neq \log_a(x^2)$. For example $\log_2(4 + 4) = 3$ and $\log_2 4 + \log_2 4 = 4$; moreover $(\log_2 8)^2 = 9$ and $\log_2 64 = 6$.

Equation $\log_a A = \log_a B$

$$\log_a A = \log_a B \iff \begin{cases} A = B \\ A > 0 \end{cases}$$

It holds for $a > 0$, $a \neq 1$, because the logarithmic function is injective. The condition $B > 0$ follows from $A = B$ and $A > 0$. If the equation is obtained with the properties from several logarithms, the conditions of existence are written for all the logarithms of the original equation.

Equation $\log_a A = k$

$$\log_a A = k \iff A = a^k$$

It holds for $a > 0$, $a \neq 1$ and k real. The condition $A > 0$ is satisfied, because $a^k > 0$.

Substitution method

$$t = \log_a x \iff x = a^t$$

It holds for $a > 0$, $a \neq 1$ and $x > 0$. The substitution transforms an equation in $\log_a x$ and $(\log_a x)^2$ into an algebraic equation in t; every real value of t is acceptable and gives the solution $x = a^t$.

Inequality $\log_a A > \log_a B$

$$\log_a A > \log_a B \iff \begin{cases} A > B \\ B > 0 \end{cases} \quad (a > 1),$$

$$\log_a A > \log_a B \iff \begin{cases} A < B \\ A > 0 \end{cases} \quad (0 < a < 1).$$

With a base greater than 1 the direction of the inequality is preserved, with a base between 0 and 1 it is reversed. In each system the condition of existence concerns the smaller argument, because the positivity of the other follows from the inequality; with $\geq$, $<$, $\leq$ the rule is the same.

Inequality $\log_a A > k$

$$\log_a A > k \iff A > a^k \quad (a > 1),$$

$$\log_a A > k \iff \begin{cases} A < a^k \\ A > 0 \end{cases} \quad (0 < a < 1).$$

For $\log_a A < k$ the two cases are exchanged, therefore the inequality is equivalent to $0 < A < a^k$ if $a > 1$ and to $A > a^k$ if $0 < a < 1$.

Exponential equations and inequalities

The examples with the two bases and with the substitution are in the chapter Exponential equations and inequalities.

Exponential function

$$y = a^x, \qquad a^x > 0 \ \text{ for every } x \in \mathbb{R}$$

It holds for $a > 0$, $a \neq 1$. The graph passes through $(0, 1)$ and has the horizontal asymptote $y = 0$; the function is strictly increasing for $a > 1$ and strictly decreasing for $0 < a < 1$. Moreover $\left(\frac{1}{a}\right)^x = a^{-x}$ holds.

Equation with the same base

$$a^A = a^B \iff A = B$$

It holds for $a > 0$, $a \neq 1$, because the exponential function is injective. Both sides are rewritten as powers of the same base, for example $8^x = 2^{3x}$ and $\frac{1}{9} = 3^{-2}$.

Equation with a constant right-hand side

$$a^A = k \iff A = \log_a k$$

It holds for $a > 0$, $a \neq 1$ and $k > 0$. For $k \leq 0$ the equation has no solutions, because $a^A > 0$.

Equation with different bases

$$a^A = b^B \iff A \log_c a = B \log_c b$$

It holds for $a, b > 0$ and for every base $c > 0$ with $c \neq 1$; it is obtained by applying $\log_c$ to both sides, which are positive. For example $2^x = 3^{x-1}$ is equivalent to $x \ln 2 = (x - 1) \ln 3$, therefore $x = \dfrac{\ln 3}{\ln 3 - \ln 2}$.

Substitution method

$$t = a^x,\ t > 0 \qquad \Longrightarrow \qquad a^{2x} = t^2, \qquad a^{x+k} = a^k\, t, \qquad a^{-x} = \frac{1}{t}$$

It holds for $a > 0$, $a \neq 1$. The solutions in t with $t \leq 0$ are discarded, and from every $t > 0$ one obtains $x = \log_a t$. In inequalities the solution set in t is intersected with $t > 0$ before returning to the variable x.

Inequalities with the same base

$$a^A > a^B \iff A > B \quad (a > 1),$$
$$a^A > a^B \iff A < B \quad (0 < a < 1).$$

With a base greater than 1 the direction of the inequality is preserved, with a base between 0 and 1 it is reversed; the same holds with $\geq$, $<$, $\leq$. For $k > 0$ one writes $k = a^{\log_a k}$, therefore for example, if $a > 1$, the inequality $a^A > k$ is equivalent to $A > \log_a k$.

Inequalities with a negative or zero right-hand side

$$a^A > k \text{ on the whole domain of } A, \qquad a^A < k \text{ never}$$

They hold for $a > 0$ and $k \leq 0$, because $a^A > 0$. Likewise $a^A \geq k$ is always satisfied and $a^A \leq k$ has no solutions.

Special values and related angles

The unit circle and its symmetries are described in the chapter Trigonometric equations and inequalities.

Degrees and radians

$$\frac{x}{\pi} = \frac{\alpha}{180^\circ}$$

Here x is the measure in radians and α the measure in degrees of the same angle. For example 30°, 45°, 60°, 90° correspond to $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, $\frac{\pi}{2}$.

Sine, cosine and tangent

$$\cos x = x_P, \qquad \sin x = y_P, \qquad \tan x = \frac{\sin x}{\cos x}$$

P is the point of the unit circle determined by the angle x, therefore $-1 \leq \sin x \leq 1$ and $-1 \leq \cos x \leq 1$. The tangent is defined for $x \neq \frac{\pi}{2} + n\pi$, the cotangent $\cot x = \frac{\cos x}{\sin x}$ for $x \neq n\pi$, with $n \in \mathbb{Z}$.

Periodicity

$$\sin(x + 2n\pi) = \sin x, \qquad \cos(x + 2n\pi) = \cos x, \qquad \tan(x + n\pi) = \tan x$$

They hold for every $n \in \mathbb{Z}$, the third where the tangent is defined. Sine and cosine have period 2π, the tangent has period π; for $k \neq 0$ the functions $\sin(kx)$ and $\cos(kx)$ have period $\frac{2\pi}{|k|}$.

Special values of sine and cosine

x	$\sin x$	$\cos x$
0	0	1
$\frac{\pi}{6}$ $(30°)$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$
$\frac{\pi}{4}$ $(45°)$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$
$\frac{\pi}{3}$ $(60°)$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$
$\frac{\pi}{2}$ $(90°)$	1	0
$\frac{2\pi}{3}$ $(120°)$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$
$\frac{3\pi}{4}$ $(135°)$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$
$\frac{5\pi}{6}$ $(150°)$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$
π $(180°)$	0	-1
$\frac{3\pi}{2}$ $(270°)$	-1	0

The angles of the third and fourth quadrants are reduced to these with the related angles, for example $\sin\frac{7\pi}{6} = -\sin\frac{\pi}{6} = -\frac{1}{2}$.

Special values of the tangent

x	$\tan x$
0	0
$\frac{\pi}{6}$	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{4}$	1
$\frac{\pi}{3}$	$\sqrt{3}$
$\frac{\pi}{2}$	undefined

In the other quadrants the values are obtained from the period π and from $\tan(-x) = -\tan x$, for example $\tan\frac{3\pi}{4} = \tan\left(-\frac{\pi}{4}\right) = -1$.

Fundamental identity and relations between the functions

$$\sin^2 x + \cos^2 x = 1, \qquad 1 + \tan^2 x = \frac{1}{\cos^2 x},$$
$$\cos^2 x = \frac{1}{1 + \tan^2 x}, \qquad \sin^2 x = \frac{\tan^2 x}{1 + \tan^2 x}.$$

The first holds for every real x, the others for $\cos x \neq 0$. Given $\sin x$, we have $\cos x = \pm\sqrt{1 - \sin^2 x}$, with the sign determined by the quadrant of the angle.

Opposite, supplementary and explementary angles, and angles differing by π

$$\begin{aligned} \sin(-x) &= -\sin x, & \cos(-x) &= \cos x \\ \sin(\pi - x) &= \sin x, & \cos(\pi - x) &= -\cos x \\ \sin(\pi + x) &= -\sin x, & \cos(\pi + x) &= -\cos x \\ \sin(2\pi - x) &= -\sin x, & \cos(2\pi - x) &= \cos x \end{aligned}$$

They hold for every real x; therefore sine and tangent are odd functions and cosine is even. For the tangent, where it is defined, $\tan(-x) = \tan(\pi - x) = \tan(2\pi - x) = -\tan x$ and $\tan(\pi + x) = \tan x$.

Complementary angles and angles differing by π/2

$$\begin{aligned} \sin\left(\frac{\pi}{2} - x\right) &= \cos x, & \cos\left(\frac{\pi}{2} - x\right) &= \sin x \\ \sin\left(\frac{\pi}{2} + x\right) &= \cos x, & \cos\left(\frac{\pi}{2} + x\right) &= -\sin x \\ \sin\left(\frac{3\pi}{2} - x\right) &= -\cos x, & \cos\left(\frac{3\pi}{2} - x\right) &= -\sin x \\ \sin\left(\frac{3\pi}{2} + x\right) &= -\cos x, & \cos\left(\frac{3\pi}{2} + x\right) &= \sin x \end{aligned}$$

They hold for every real x. Where both sides are defined, $\tan\left(\frac{\pi}{2} - x\right) = \frac{1}{\tan x}$ and $\tan\left(\frac{\pi}{2} + x\right) = -\frac{1}{\tan x}$.

Trigonometric identities

The addition and double-angle formulas are stated in the chapter Trigonometric equations and inequalities; the others are derived from these.

Addition and subtraction formulas

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b,$$

$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b,$$

$$\tan(a \pm b) = \frac{\tan a \pm \tan b}{1 \mp \tan a \tan b}.$$

The first two hold for all real a, b, the third where $\tan a$, $\tan b$ and $\tan(a \pm b)$ are defined. For example $\cos \dfrac{\pi}{12} = \cos\left(\dfrac{\pi}{3} - \dfrac{\pi}{4}\right) = \dfrac{\sqrt{2}+\sqrt{6}}{4}$.

Double-angle formulas

$$\sin 2x = 2 \sin x \cos x,$$

$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1,$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}.$$

The first two hold for every real x, the third where $\tan x$ and $\tan 2x$ are defined. With the suitable form of $\cos 2x$ an equation containing $\cos 2x$ and $\sin x$, or $\cos 2x$ and $\cos x$, becomes a quadratic equation in a single function.

Triple-angle formulas

$$\sin 3x = 3 \sin x - 4 \sin^3 x, \qquad \cos 3x = 4 \cos^3 x - 3 \cos x$$

They hold for every real x and are obtained by writing $3x = 2x + x$ and applying the addition and double-angle formulas.

Power-reduction formulas

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}, \qquad \sin x \cos x = \frac{\sin 2x}{2}$$

They hold for every real x and follow from the double-angle formulas. They transform a second-degree expression in sine and cosine into a first-degree expression in $\sin 2x$ and $\cos 2x$.

Half-angle formulas

$$\sin\frac{x}{2} = \pm\sqrt{\frac{1-\cos x}{2}}, \qquad \cos\frac{x}{2} = \pm\sqrt{\frac{1+\cos x}{2}},$$
$$\tan\frac{x}{2} = \frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}.$$

In the first two the sign is that of the function in the quadrant where $\frac{x}{2}$ lies. The two expressions of $\tan\frac{x}{2}$ hold respectively for $\cos x \neq -1$ and for $\sin x \neq 0$.

Parametric formulas

$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \tan x = \frac{2t}{1-t^2},$$
$$t = \tan\frac{x}{2}.$$

They hold for $x \neq \pi + 2n\pi$, where $\tan\frac{x}{2}$ is defined, and the third also requires $t \neq \pm 1$. With this substitution the equation $a\sin x + b\cos x = c$ becomes a quadratic equation in t; the angles $x = \pi + 2n\pi$ are checked separately in the original equation.

Sum-to-product formulas

$$\sin p + \sin q = 2\sin\frac{p+q}{2}\cos\frac{p-q}{2}$$
$$\sin p - \sin q = 2\cos\frac{p+q}{2}\sin\frac{p-q}{2}$$
$$\cos p + \cos q = 2\cos\frac{p+q}{2}\cos\frac{p-q}{2}$$
$$\cos p - \cos q = -2\sin\frac{p+q}{2}\sin\frac{p-q}{2}$$

They hold for all real p, q. They transform a sum into a product, to which the zero product property is applied.

Product-to-sum formulas

$$\sin a\sin b = \frac{\cos(a-b) - \cos(a+b)}{2}$$
$$\cos a\cos b = \frac{\cos(a-b) + \cos(a+b)}{2}$$
$$\sin a\cos b = \frac{\sin(a+b) + \sin(a-b)}{2}$$

They hold for all real a, b and transform a product into a sum.

Trigonometric equations and inequalities

The solving methods, with an example for each type of equation, are in the chapter Trigonometric equations and inequalities.

Elementary equations

$$\begin{aligned}\sin x = k &\iff x = \alpha + 2n\pi \ \vee \ x = \pi - \alpha + 2n\pi \\ \cos x = k &\iff x = \alpha + 2n\pi \ \vee \ x = -\alpha + 2n\pi \\ \tan x = k &\iff x = \alpha + n\pi\end{aligned}$$

In each row α is an angle with $\sin\alpha = k$, $\cos\alpha = k$, $\tan\alpha = k$ respectively, and $n \in \mathbb{Z}$. The first two hold for $|k| \leq 1$, and for $|k| > 1$ the equations have no solutions; the third holds for every real k.

Arcsine, arccosine and arctangent

$$\begin{aligned}\alpha = \arcsin k &\in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \\ \alpha = \arccos k &\in [0, \pi] \\ \alpha = \arctan k &\in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\end{aligned}$$

These are the angles used as α when k is not a special value; the first two exist for $|k| \leq 1$, the arctangent for every real k. For example the solutions of $\tan x = 2$ are $x = \arctan 2 + n\pi$.

Particular cases

$$\begin{aligned}\sin x = 0 &\iff x = n\pi, & \cos x = 0 &\iff x = \frac{\pi}{2} + n\pi \\ \sin x = 1 &\iff x = \frac{\pi}{2} + 2n\pi, & \cos x = 1 &\iff x = 2n\pi \\ \sin x = -1 &\iff x = -\frac{\pi}{2} + 2n\pi, & \cos x = -1 &\iff x = \pi + 2n\pi\end{aligned}$$

They hold with $n \in \mathbb{Z}$. In these cases the two families of solutions of the elementary equations reduce to one.

Equality of two sines, of two cosines, of two tangents

$$\begin{aligned}\sin A = \sin B &\iff A = B + 2n\pi \ \vee \ A = \pi - B + 2n\pi \\ \cos A = \cos B &\iff A = B + 2n\pi \ \vee \ A = -B + 2n\pi \\ \tan A = \tan B &\iff A = B + n\pi\end{aligned}$$

They hold with $n \in \mathbb{Z}$, the third where $\tan A$ and $\tan B$ are defined. The equation $\sin A = \cos B$ is reduced to the first by writing $\cos B = \sin\left(\frac{\pi}{2} - B\right)$.

Quadratic equations in sine or cosine

$$a\sin^2 x + b\sin x + c = 0 \quad \xrightarrow{t = \sin x} \quad \begin{cases} at^2 + bt + c = 0 \\ -1 \le t \le 1 \end{cases}$$

Every root t with $|t| \le 1$ gives the elementary equation $\sin x = t$, and the roots with $|t| > 1$ are discarded; the same holds with $t = \cos x$. If $\sin^2 x$ and $\cos x$ appear, the substitution $\sin^2 x = 1 - \cos^2 x$ brings the equation to a single function.

Homogeneous quadratic equations

$$a\sin^2 x + b\sin x\cos x + c\cos^2 x = 0 \iff a\tan^2 x + b\tan x + c = 0$$

It holds for $a \neq 0$, because in this case the points with $\cos x = 0$ are not solutions and the division by $\cos^2 x$ loses no solutions. For $a = 0$ one factors out $\cos x$. With a constant term d on the right-hand side one writes $d = d\,(\sin^2 x + \cos^2 x)$ and the equation becomes homogeneous.

Auxiliary angle method

$$a\sin x + b\cos x = R\sin(x + \varphi), \qquad R = \sqrt{a^2 + b^2},$$

$$\begin{cases} \cos\varphi = \dfrac{a}{R} \\ \sin\varphi = \dfrac{b}{R} \end{cases}$$

It holds for a and b not both zero. The linear equation $a\sin x + b\cos x = c$ becomes $\sin(x + \varphi) = \dfrac{c}{R}$, which has solutions *if and only if* $|c| \le R$.

Warning

Dividing by a factor that can vanish loses solutions. In $\sin x\cos x = \cos x$ factoring gives $\cos x\,(\sin x - 1) = 0$, with solutions $x = \dfrac{\pi}{2} + n\pi$; dividing by $\cos x$ would leave only $\sin x = 1$, that is, $x = \dfrac{\pi}{2} + 2n\pi$.

Elementary inequalities

$$\begin{aligned}
\sin x > k &\iff \alpha + 2n\pi < x < \pi - \alpha + 2n\pi \\
\sin x < k &\iff \pi - \alpha + 2n\pi < x < 2\pi + \alpha + 2n\pi \\
\cos x > k &\iff -\beta + 2n\pi < x < \beta + 2n\pi \\
\cos x < k &\iff \beta + 2n\pi < x < 2\pi - \beta + 2n\pi \\
\tan x > m &\iff \gamma + n\pi < x < \frac{\pi}{2} + n\pi
\end{aligned}$$

The first four hold for $|k| < 1$, with $\alpha = \arcsin k$ and $\beta = \arccos k$; the last holds for every real m, with $\gamma = \arctan m$, and $n \in \mathbb{Z}$. For $k \geq 1$ the inequality $\sin x > k$ has no solutions, for $k < -1$ it is satisfied by every x and for $k = -1$ by every x with $\sin x \neq -1$; the same holds for the cosine.

Lines and the Cartesian plane

The formulas for the line are derived in the chapter Analytic geometry and conic sections.

Distance between two points and midpoint

$$\overline{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \qquad M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

They hold for $A(x_1, y_1)$ and $B(x_2, y_2)$. The distance is the Pythagorean theorem applied to the triangle with legs $|x_2 - x_1|$ and $|y_2 - y_1|$.

Equation of a line

$$y = mx + q, \qquad ax + by + c = 0, \qquad m = -\frac{a}{b}, \quad q = -\frac{c}{b}$$

The slope-intercept form describes the non-vertical lines, the general form, with $(a, b) \neq (0, 0)$, describes every line; the passage from one to the other holds for $b \neq 0$. Vertical lines have equation $x = k$, horizontal lines $y = k$.

Slope and line through a point

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \qquad y - y_0 = m\,(x - x_0)$$

The first is the slope of the line through $A(x_1, y_1)$ and $B(x_2, y_2)$ and holds for $x_1 \neq x_2$. The second is the line with slope m passing through $P(x_0, y_0)$; as m varies it describes the pencil of non-vertical lines through P.

Line through two points

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

It holds for $x_1 \neq x_2$ and $y_1 \neq y_2$. If $x_1 = x_2$ the line is $x = x_1$, if $y_1 = y_2$ it is $y = y_1$.

Parallel and perpendicular lines

$$r \parallel s \iff m_1 = m_2, \qquad r \perp s \iff m_1 m_2 = -1$$

They hold for non-vertical lines with slopes m_1 and m_2; the perpendicular line therefore has slope $-\dfrac{1}{m_1}$. In general form the lines $ax + by + c = 0$ and $a'x + b'y + c' = 0$ are parallel if and only if $ab' - a'b = 0$ and perpendicular if and only if $aa' + bb' = 0$.

Distance from a point to a line

$$d(P, r) = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

It holds for the point $P(x_0, y_0)$ and the line r written in general form $ax + by + c = 0$.

Intersection of two lines

$$\begin{cases} ax + by + c = 0 \\ a'x + b'y + c' = 0 \end{cases}$$

The solutions of the system are the coordinates of the points common to the two lines. The system has exactly one solution if $ab' - a'b \neq 0$, none if the lines are parallel and distinct, infinitely many if they coincide.

Conic sections

The definitions as loci and the procedure for reduction to standard form are in the chapter Analytic geometry and conic sections.

Parabola with vertical axis

$$y = ax^2 + bx + c, \qquad V\left(-\frac{b}{2a}, -\frac{\Delta}{4a}\right), \qquad x = -\frac{b}{2a}$$

It holds for $a \neq 0$, with $\Delta = b^2 - 4ac$; the line $x = -\dfrac{b}{2a}$ is the axis of symmetry, and the ordinate of the vertex is also obtained by substituting x_V into the equation. The parabola opens upward for $a > 0$ and downward for $a < 0$; it meets the y-axis at $(0, c)$ and the x-axis at the solutions of $ax^2 + bx + c = 0$.

Focus and directrix of the parabola

$$F\left(-\frac{b}{2a}, \frac{1-\Delta}{4a}\right), \qquad y = -\frac{1+\Delta}{4a}$$

They hold for the parabola $y = ax^2 + bx + c$ with $a \neq 0$. The parabola is the locus of the points equidistant from the focus F and from the directrix.

Parabola with horizontal axis

$$x = ay^2 + by + c, \qquad V\left(-\frac{\Delta}{4a}, -\frac{b}{2a}\right), \qquad y = -\frac{b}{2a}$$

It holds for $a \neq 0$, with $\Delta = b^2 - 4ac$. It is obtained from the parabola with vertical axis by exchanging x and y; the parabola opens to the right for $a > 0$ and to the left for $a < 0$.

Circle

$$(x - x_0)^2 + (y - y_0)^2 = r^2, \qquad x^2 + y^2 + ax + by + c = 0$$

The first is the circle with center $C(x_0, y_0)$ and radius $r > 0$. Expanding the squares one obtains the second, with $a = -2x_0$, $b = -2y_0$ and $c = x_0^2 + y_0^2 - r^2$.

Center and radius from the general form

$$C\left(-\frac{a}{2}, -\frac{b}{2}\right), \qquad r = \sqrt{\frac{a^2}{4} + \frac{b^2}{4} - c}$$

The equation $x^2 + y^2 + ax + by + c = 0$ represents a circle *only if* $\frac{a^2}{4} + \frac{b^2}{4} - c > 0$. If the quantity is zero the equation represents only the point C, if it is negative no point.

Completing the square

$$x^2 + kx = \left(x + \frac{k}{2}\right)^2 - \frac{k^2}{4}, \qquad Ax^2 + Bx = A\left(x + \frac{B}{2A}\right)^2 - \frac{B^2}{4A}$$

The first holds for every real k, the second for $A \neq 0$. By completing the square in x and in y a second-degree equation without the xy term is brought to standard form.

Position of a line relative to a circle

$$\begin{aligned} d(C, s) < r &\iff s \text{ is a secant} \\ d(C, s) = r &\iff s \text{ is a tangent} \\ d(C, s) > r &\iff s \text{ is external} \end{aligned}$$

Here $d(C, s)$ is the distance from the center C to the line s, computed with the formula for the distance from a point to a line, and r is the radius.

Ellipse

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, \qquad c^2=a^2-b^2 \quad (a>b)$$

It holds for $a, b>0$. The vertices are $(\pm a, 0)$ and $(0, \pm b)$; if $a>b$ the foci are $(\pm c, 0)$, if $b>a$ they are $(0, \pm c)$ with $c^2=b^2-a^2$. The eccentricity is the ratio of c to the semi-major axis and is less than 1.

Hyperbola

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1, \qquad c^2=a^2+b^2, \qquad y=\pm\frac{b}{a}x$$

It holds for $a, b>0$. The vertices are $(\pm a, 0)$, the foci $(\pm c, 0)$ and the lines $y=\pm\frac{b}{a}x$ are the asymptotes. The equation $\frac{x^2}{a^2}-\frac{y^2}{b^2}=-1$ represents the hyperbola with vertices $(0, \pm b)$, foci $(0, \pm c)$ and the same asymptotes.

Rectangular hyperbola referred to its asymptotes

$$xy=k$$

It holds for $k \neq 0$, and the asymptotes are the coordinate axes. For $k>0$ the branches lie in the first and third quadrants, with vertices $(\sqrt{k}, \sqrt{k})$ and $(-\sqrt{k}, -\sqrt{k})$; for $k<0$ they lie in the second and fourth.

Homographic function

$$y=\frac{ax+b}{cx+d}, \qquad x=-\frac{d}{c}, \qquad y=\frac{a}{c}$$

It holds for $c \neq 0$ and $ad-bc \neq 0$. The graph is a rectangular hyperbola with vertical asymptote $x=-\frac{d}{c}$ and horizontal asymptote $y=\frac{a}{c}$; for $ad-bc=0$ the function is constant where it is defined.

Identifying a conic section

$$Ax^2+Cy^2+Dx+Ey+F=0$$

- exactly one of A and C is zero and the other variable appears to the first degree: parabola;
- $A=C \neq 0$: circle;
- A and C with the same sign and different: ellipse;
- A and C with opposite signs: hyperbola.

Completing the squares confirms the type and can give a degenerate case: a point, no point or a pair of lines.

Translated ellipse and hyperbola

$$\frac{(x-x_0)^2}{a^2}+\frac{(y-y_0)^2}{b^2}=1, \qquad \frac{(x-x_0)^2}{a^2}-\frac{(y-y_0)^2}{b^2}=1$$

They are the standard ellipse and hyperbola translated by the vector (x_0, y_0), therefore center, vertices, foci and asymptotes are obtained from the standard ones by adding x_0 to the abscissas and y_0 to the ordinates.

Position of a line relative to a conic section

$$\begin{cases} y = mx + q \\ f(x,y) = 0 \end{cases} \quad \longrightarrow \quad \alpha x^2 + \beta x + \gamma = 0$$

Substituting the equation of the line into that of the conic $f(x,y)=0$ one obtains the resolvent equation; if it is of second degree, the line is a secant for $\Delta > 0$, a tangent for $\Delta = 0$ and external for $\Delta < 0$. The tangent lines from a point $P(x_0, y_0)$ are found by imposing $\Delta = 0$ on the line $y - y_0 = m(x - x_0)$; the vertical line $x = x_0$ is checked separately.